Patient Robot Advocacy

A Phase 1, First-in-Human, PDAC Clinical Trial Protocol of a LLM-Directed Robotic Whipple with Daraxonrasib (RMC-6236)

Draft 1.0 | **Repository v1.0.0** | **30 figures** | **21 documented concerns answered**

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10.5281/zenodo.xxxxxxxx | 0009-0007-5457-8667

Who is in control?
A named human surgeon approves every motion. The model has no path to the arms. Stopping is bounded at 3 ms across the arms and 500 ms system-wide. §7, Figures 18 and 19.

What could go wrong?
Twenty-one documented concerns, each wired to the clause, limit, or number that answers it, including the five that are answered only by governance. §3, Figures 5, 6, and 7.

What is this costing me?
Roughly 27 days of contact over 24 months, 18 visits, one inpatient stay. The sponsor pays study-only items; two items remain unsettled. §9 and §12, Figures 25 and 30.

Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Disclaimer: *This work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.*

San Diego
July 31, 2026

The paper (v1.0) is at https://doi.org/10.5281/zenodo.xxxxxxxx and the repository (v1.0.0) is at https://github.com/kevinkawchak/robotic-surgeries/tree/main/patient-robot-advocacy. The parent clinical protocol this paper advocates for is 10.5281/zenodo.20780121 [1]; bill citations use H. R. 9510 v5, 10.5281/zenodo.20619762 [2].

Keywords Patient advocacy; Informed consent; Pancreatic Ductal Adenocarcinoma; Robotic Whipple; Daraxonrasib; Physical AI; On-premises LLM; Surgical autonomy; Patient concerns; Phase 1 Trial Protocol; H. R. 9510

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[DRAFTING INSTRUCTION] *This is the draft stage. Every bracketed instruction below names the exact file in `patient-robot-advocacy/` that the `full-patient` stage must read to replace it. No instruction* *may survive into* `full-patient/`*. The figure sources are in* `mermaid/`, `plantuml/`, `d2/`, `diagrams-python/`*, and* `graphviz/`*; the concern evidence is in* `research/research-a.md` *and* `research/research-b.md`*; the clinical facts are in* `inputs/phase-1-trial-protocol.zip`*; the bill citation is* `references/references.bib` *key* `hr9510billv5`*.*

1 Statement of Patient Commitment

The parent clinical protocol opens with a Statement of Compliance addressed to the Food and Drug Administration, the Institutional Review Board, and the sponsor [1]. That statement is necessary and it is not addressed to the person whose abdomen will be opened. This paper opens instead with the seven things the same protocol commits to the participant, each one tied to the clause that makes it enforceable rather than aspirational.

The premise is taken from the author's earlier analysis of proposed United States legislation for Physical AI oncology trials: the cancer patient, not the doctor, the nurse, the trial sponsor, the Institutional Review Board, or the regulator, is the priority participant [3, 2].

[DRAFTING INSTRUCTION] *Expand this opening to roughly 1,300 characters. Source the framing from* `inputs/patient-priority-physical-ai.zip` *and cite* `hr9510billv5` *for the legislative premise. State explicitly that this is an advocacy document written from the protocol's own record, not a substitute for the consent form.*

1.1 The seven commitments

[DRAFTING INSTRUCTION] *Write one short paragraph per commitment, roughly 420 characters each, in the order below. Each paragraph names the commitment, states the protocol clause that carries it, and states what a participant can check to verify it. Source every clause from* `inputs/phase-1-trial-protocol.zip`*, sections* `sec-00-compliance`*,* `sec-04-design`*, and* `sec-06-intervention`*.*

#	Commitment	The clause that makes it enforceable
1	Nothing starts without you	Consent under 21 CFR §50.20 precedes every study procedure
2	The robot is proven before you	Phase 0 gate: at least 1000 simulations across at least two independent frameworks, Unified Safety Level at least 7.0
3	You are never the untested dose	3+3 escalation with sentinel staggering; a dose level opens only after the level below is cleared
4	A human approves every motion	Class II collaborative device under continuous human oversight; no model-to-actuator path exists
5	Stopping is faster than harm	Emergency stop at most 3 ms across arms and at most 500 ms system-wide
6	Your record cannot be edited	Hash-chained audit trail under 21 CFR part 11, bound to a deterministic seed and a repository commit
7	Leaving costs you nothing	Withdrawal at any visit without reason; standard care continues unchanged

[DRAFTING INSTRUCTION] *Table above is final in structure. Verify the column widths sum to* `textwidth`*: 0.7cm plus 4.6cm plus the Y remainder, with* `tabcolsep` *5pt on six edges.*

1.2 What this paper is, and what it is not

[DRAFTING INSTRUCTION] *Roughly 900 characters. It is an independent advocacy paper and a practical adoption guide built from the author's repository sources. It is not the consent form, not medical advice, not regulatory advice, and not endorsed by any sponsor, CRO, site, IRB, regulator, or medical society. Cite* `NCIConsent24` *for what a consent form is required to contain, and state that this paper is designed to be read before that form, not instead of it.*

1.3 How to read the thirty figures

[DRAFTING INSTRUCTION] *Roughly 800 characters. Explain the five diagram types and the question each answers, drawing the wording from the five stage READMEs at* `mermaid/README.md`*,* `plantuml/README.md`*,* `d2/README.md`*,* `diagrams-python/README.md`*, and* `graphviz/README.md`*. State that the monospace tag at the top left of every frame names the native construct the figure reproduces, so a reader can carry any figure back to its platform.*

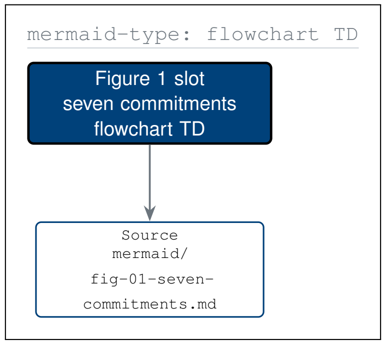

Figure 1. The seven commitments made to the participant, from the first conversation through consent, the Phase 0 gate, and dose assignment, to the day-30 safety window and the day-90 pathology result that is reported back to them directly.

[DRAFTING INSTRUCTION] *Replace the Figure 1 slot with the full mermaid-type flowchart specified in* `mermaid/fig-01-seven-commitments.md`*, following the TikZ rendering notes at the foot of that file: seven* `mmin` *commitment nodes in one column at 2.4 cm pitch, three* `mmdec` *gates at $x = 5.6$, one* `mmdark` *halt node reached by three* `mmedged` *edges, and the day-30 and day-90 nodes below.*

1.4 Accountability, stated once, at the front

[DRAFTING INSTRUCTION] *Roughly 700 characters. The single most common accountability question in the surveyed literature is "who is to blame?" [4, 5]. Answer it here in summary and forward to §11: there is exactly one accountable party per failure class, named in the responsibility matrix of Figure 29, and the participant is told which one to call.*

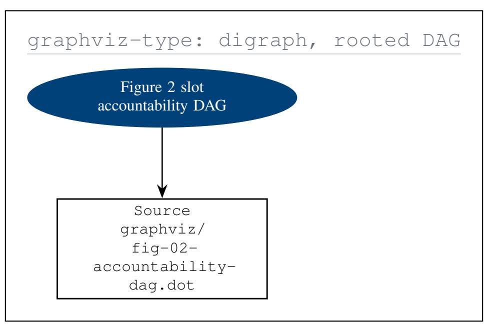

Figure 2. The accountability chain drawn as a rooted directed acyclic graph with the participant at the root, showing the people in the room, the bodies accountable for the study, the independent reviewers, and the three obligations that point back.

[DRAFTING INSTRUCTION] *Replace the Figure 2 slot with the full graphviz-type DAG specified in* `graphviz/fig-02-accountability-dag.dot`*. Preserve the stated invariant: the graph is acyclic, so no body in the chart is its own reviewer. Draw exactly three back-reference edges to the root, no more, because three is what the protocol actually owes the participant.*

2 Plain-Language Protocol Summary

2.1 The study in one paragraph

[DRAFTING INSTRUCTION] *Roughly 1,100 characters, written at the reading level of a well-informed adult with no clinical training. Adapt the synopsis from `inputs/phase-1-trial-protocol.zip`, `sections/sec-01-summary.tex` §1.1, and remove every term that a consent conversation would have to define twice. Cite `Siegel2025` for the survival baseline and `rasolute302` for the drug. State up front that this is a first-in-human study of a combination, and that the primary question is safety, not cure.*

2.2 What is being tested, in three parts

[DRAFTING INSTRUCTION] *Three short paragraphs, roughly 500 characters each. Part one: the drug, daraxonrasib, a RAS(ON) multi-selective inhibitor, given before and after surgery, cite `rasolute302` and `oreilly2026darax`. Part two: the operation, a pancreaticoduodenectomy performed with an eight-arm robotic platform, cite `onpremwhippl` and `DutchCohort2025`. Part three: the software, an on-premises large language model that advises and never acts, cite `LeeAuto2024X` for where that sits on the published autonomy scale.*

2.3 Your journey, start to finish

[DRAFTING INSTRUCTION] *Roughly 900 characters introducing Figure 3, and stating explicitly that the journey structure is adapted from the author's autonomous single-patient journey simulation [6], which was non-small cell lung cancer, and that this study is PDAC. Name the four differences that matter: the operation, the drug, the dominant early complication, and the survival baseline. Source the distinction from `inputs/cancer-patient-journey.zip` and `inputs/README.md`.*

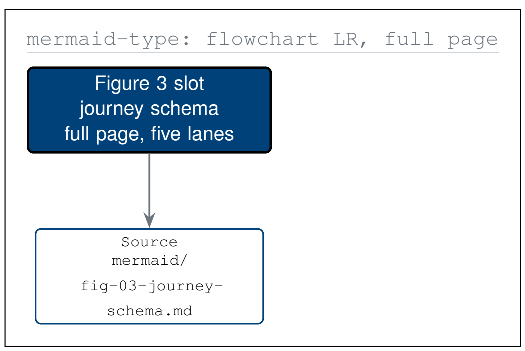

Figure 3. The participant journey across five phases, screening through the 24-month endpoint, with every node the participant controls filled Corporate Blue and every node the sponsor controls left white, so the balance of control is visible.

[DRAFTING INSTRUCTION] *Replace the Figure 3 slot with the full-page mermaid-type flowchart from `mermaid/fig-03-journey-schema.md`. Five `mmlane` lanes at $x = 0, 4.6, 9.2, 13.4, 18.0$; four participant `mmgoal` nodes; the operation in `pablue1`; three dotted exits routed below all lanes at $y = -11.2$ with right-angle routing, never diagonally through a lane.*

2.4 Your schedule, and what it costs you in time

[DRAFTING INSTRUCTION] *Roughly 800 characters introducing Figure 4 and the totals it carries. Adapt from the Schedule of Activities in `inputs/phase-1-trial-protocol.zip`, `sections/sec-01-summary.tex` §1.3, and convert every cross in that matrix into a statement of what the participant does. Cite `NCITrial2024` for the general shape of trial participation.*

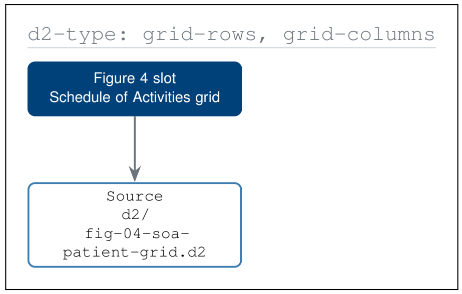

*Figure 4. The Schedule of Activities redrawn as a true grid from the participant's chair, with each cell stating what they do rather than what is collected,
and a totals row giving the approximate hours of their time each visit consumes.*

[DRAFTING INSTRUCTION] *Replace the Figure 4 slot with the full D2-type grid from* `d2/fig-04-soa-patient-grid.d2`*: eight columns by thirteen rows of* `d2cell` *rectangles sharing their rules, header row* `d2cellh`*, label column* `d2celll`*, participant-decision cells* `protoblue`*, activity cells* `pablue1`*, totals row* `pagray1`*. Add the four-key legend beneath.*

2.5 Schedule of Activities, in the conventional form

[DRAFTING INSTRUCTION] *Retain the conventional matrix as well as the grid, because a participant may want to hand it to their own oncologist. Populate every cell from* `inputs/phase-1-trial-protocol.zip`*,* `sections/sec-01-summary.tex` *§1.3, unchanged.*

Activity	Screen -28 to -1	Base Day -1	Surg Day 0	Acute Day 1-7	Day 30	Day 90	LT q12wk
Informed consent, including Physical AI opt-out	X						
Staging and KRAS G12 confirmation	X						
CA 19-9	X	X			X	X	X
ECOG performance status	X	X			X	X	X
Safety laboratory panel	X	X	X	X	X	X	X
Phase 0 simulation sign-off		X					
USL at least 7.0 and safety matrix		X					
Robotic Whipple surgery			X				
Intra-operative telemetry capture			X				
ISGPS fistula grading				X	X		
Daraxonrasib restart advisory				X			
Adverse event review	X	X	X	X	X	X	X
RECIST imaging	X					X	X
PFS and OS assessment						X	X

Abbreviations: ISGPS, International Study Group on Pancreatic Surgery; LT, long-term; OS, overall survival; PFS, progression-free survival; RECIST, Response Evaluation Criteria in Solid Tumors; USL, Unified Safety Level.

2.6 What the study cannot tell you

[DRAFTING INSTRUCTION] *Roughly 700 characters, and do not soften it. A Phase 1 study of at most eighteen participants is powered for safety and feasibility and for nothing else. It cannot establish that this combination improves survival. Cite `NCITrial2024` and `FDARobot2022`, the latter for the FDA's own statement that long-term cancer outcomes have not been established for every use of robotically assisted surgical devices. This subsection is load-bearing for the paper's credibility and must not be trimmed in `full-patient`.*

3 The Documented Patient Concerns

3.1 Where the twenty-one come from

[DRAFTING INSTRUCTION] *Roughly 1,000 characters. Two dated research passes on July 28, 2026, one by Gemini 3.1 Pro and one by ChatGPT 5.6 Thinking Extended, filed at `research/research-a.md` and `research/research-b.md`. Six concern families and sixteen numbered concerns, overlapping on five items, de-duplicating to twenty-one. Cite the underlying human evidence: `WuSemiAI2026`, `TelesAI2026A`, `Jauniaux2025`, and `BrarRAS2024X`. State the two headline numbers: 56 percent of 50 previously operated patients would consider semi-autonomous robot-assisted surgery, conditional on which steps the robot controls; and approximately half of 330 oncology patients expressed concern about artificial intelligence in cancer care.*

Concern family	Concerns	Where this paper answers it
Control and the human element	5	§7.3, §7.4; Figures 8, 18, 19
Safety and technical failure	5	§7.4, §7.5, §7.6; Figures 20, 21
Evidence and effectiveness	4	§4.2, §5.1, §10; Figures 13, 27, 28
Data, privacy, and security	3	§9.3, §9.4; Figures 17, 24
Fairness and applicability	2	§6.3, §8.3; Figures 14, 23
Accountability and burden	2	§11, §12; Figures 29, 30

[DRAFTING INSTRUCTION] *Verify the concern count sums to 21 and that every family row names at least one section and one figure. Source the mapping from `research/README.md`, which carries both concern-to-section tables.*

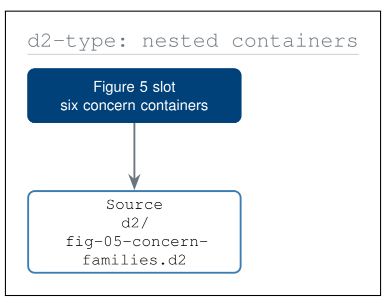

Figure 5. The twenty-one documented concerns grouped into six disjoint families, with the answering section and figure printed inside each item, and a fill key stating how many are answered by a limit, a guarantee, or governance alone.

[DRAFTING INSTRUCTION] *Replace the Figure 5 slot with the full D2-type nested-container figure from `d2/fig-05-concern-families.d2`. Six `d2cont` containers in a 3 by 2 arrangement at the coordinates given in that file's TikZ notes. Fill key: seven items Corporate Blue for a hard numeric limit, nine white for a procedural guarantee, five `pagraym` for governance or disclosure only.*

3.2 The five families, one subsection each

3.2.1 Control and the human element

[DRAFTING INSTRUCTION] *Roughly 1,400 characters. Concerns 1 to 5: who is driving, can the surgeon stop it, will the surgeon defer to the model, will I still have a doctor who knows me, less time with my care team. Cite `WuSemiAI2026` and `BrarRAS2024X`; the latter reports that the most common public misconception is that the robot operates by itself. Answer with the Class II collaborative classification and the absence of a model-to-actuator path, and forward to Figures 8 and 19.*

3.2.2 *Safety and technical failure*

[DRAFTING INSTRUCTION] *Roughly 1,400 characters. Concerns 6 to 10: malfunction or unintended motion, misreading tissue or a margin, injury to a major vessel, instrument failure mid-procedure, conversion to open surgery. Answer with the force caps, the vascular no-fly corridor, and the fault tree, and state plainly that conversion to open surgery is a recorded outcome rather than a failure. Cite* `iec8060127` *and* `ieee7009` *for the standards the safety architecture is built against.*

3.2.3 *Evidence and effectiveness*

[DRAFTING INSTRUCTION] *Roughly 1,400 characters. Concerns 11 to 14: cancer control, being among the first, newer described as better, and team experience. Cite* `DutchCohort2025` *for the learning-curve comparator,* `Jauniaux2025` *for the systematic divergence between expectation and experience, and* `FDARobot2022` *for the FDA's own caution. This is the family the paper answers least completely, and §3.4 must say so.*

3.2.4 *Data, privacy, and security*

[DRAFTING INSTRUCTION] *Roughly 1,200 characters. Concerns 15 to 17: who sees the recording, will my data train other systems, could the system be attacked. Answer with the on-premises boundary, the per-store read lists, and the retention periods. Cite* `FDACyber2026` *and* `FDATrans2024`*, and forward to Figures 17 and 24.*

3.2.5 *Fairness, applicability, accountability, and burden*

[DRAFTING INSTRUCTION] *Roughly 1,200 characters. Concerns 18 to 21: tested on people like me, software change after consent, who is accountable, and what it costs. Cite* `FDADiver2017` *for subgroup reporting and* `FDARealW2025` *for real-world performance monitoring. Forward to Figures 14, 23, 29, and 30.*

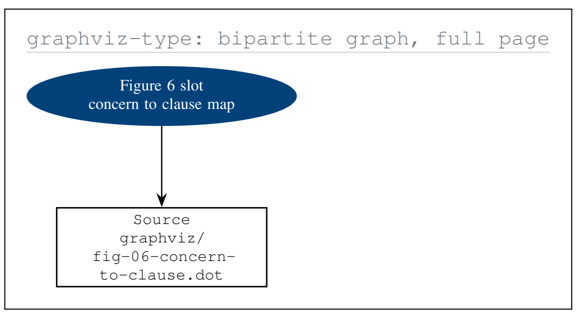

Figure 6. Every documented concern wired to the clause, limit, or number that answers it, drawn as a strictly bipartite graph so that no concern is answered with another concern, and dashed where the answer is governance or disclosure alone.

[DRAFTING INSTRUCTION] *Replace the Figure 6 slot with the full-page graphviz-type bipartite map from* `graphviz/fig-06-concern-to-clause.dot`*. Twenty-one concern ellipses on the left, nineteen clause boxes on the right, twenty-four edges between them and none within a partition. Fourteen edges solid Corporate Blue, seven solid gray, three dashed. This is the load-bearing figure of the paper.*

3.3 How completely each concern is answered

[DRAFTING INSTRUCTION] *Roughly 900 characters introducing Figure 7 and its two axes: prevalence in the surveyed sources, and completeness of the protocol answer scored 1.0 for a numeric limit, 0.7 for a procedural guarantee, 0.5 for a governance commitment, and 0.3 for disclosure only. Name the four concerns in quadrant 4 and do not argue them away.*

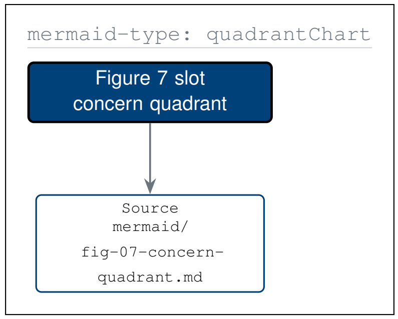

*Figure 7. Sixteen documented concerns positioned by how often they are raised in the surveyed literature against how completely this protocol answers them,
with the lower-right quadrant collecting the concerns answered by governance alone.*

[DRAFTING INSTRUCTION] *Replace the Figure 7 slot with the mermaid-type quadrant from `mermaid/fig-07-concern-quadrant.md`. Fill the four quadrant fields first, then the dividing cross, then the sixteen points as 1.6 mm Corporate Blue discs. Where two points fall within 8 mm, alternate the label anchor and add a 0.4pt leader so no label overlaps another.*

3.4 The same safeguards, read as capabilities you hold

[DRAFTING INSTRUCTION] *Roughly 800 characters introducing Figure 8. Every safeguard in the parent protocol is written with the sponsor as the actor. Reassigning the primary actor is the intervention. Ten of the twelve capabilities are exercised by the participant; the two that are not require someone physically present or inside the build system, and neither can be exercised against the participant.*

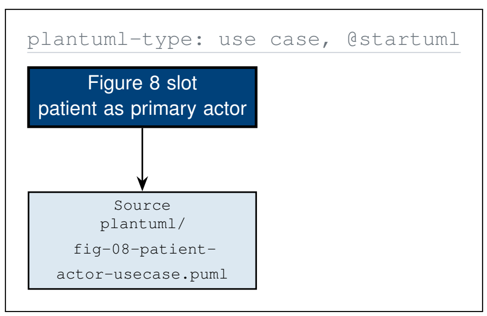

*Figure 8. The twelve safeguards of the protocol redrawn as a use case diagram with the participant as the primary actor, so that each one reads as a capability they
hold rather than as an obligation the sponsor has undertaken on their behalf.*

[DRAFTING INSTRUCTION] *Replace the Figure 8 slot with the PlantUML-type use case diagram from `plantuml/fig-08-patient-actor-usecase.puml`. Participant actor at $(-6.4, -4.2)$, left of the boundary; three secondary actors right of it. Twelve `umlusecase` ellipses in three columns and four rows. The two long associations to the far column route with looseness 0.85 so they pass between rows rather than through an ellipse.*

3.5 Where each concern physically lives

[DRAFTING INSTRUCTION] *Roughly 700 characters introducing Figure 9. A concern with no physical address is unanswerable; a concern that resolves to a named cabinet, a named console, or a named cable can be inspected. Source from `diagrams-python/fig_09_concern_locations.py` and its `CONCERN_ADDRESS` mapping.*

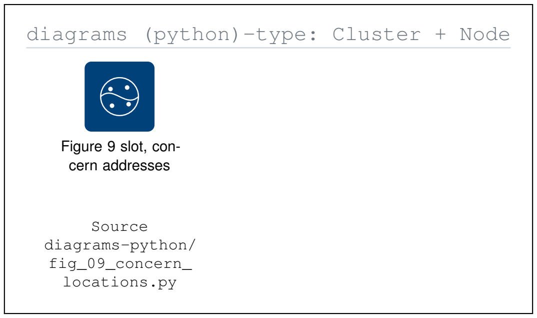

Figure 9. Each documented concern resolved to the physical component that answers it, grouped into the operating room, the on-premises cabinet, the governance bodies, and everything outside the building, with the intra-procedural route absent.

[DRAFTING INSTRUCTION] *Replace the Figure 9 slot with the full Diagrams (Python)-type clustered map from `diagrams-python/fig_09_concern_locations.py`. Four `dgcluster` containers, thirteen concern badges anchored north east on their tiles with a 1 mm outset, and the single cut edge terminated by a `\pxmark` cross labelled "no route exists during a procedure".*

4 Objectives and Patient-Facing Endpoints

4.1 What the study is trying to find out

[DRAFTING INSTRUCTION] *Roughly 1,100 characters. Adapt the objectives from `inputs/phase-1-trial-protocol.zip`, `sections/sec-03-objectives.tex`, and restate each as a question rather than a noun phrase. The primary objective asks whether the combination is safe and feasible; the secondary objectives ask how the operation and the drug performed; the exploratory objective asks what the advisory layer contributed. Cite `onpremwhippl`.*

4.2 Every endpoint, and the sentence it will let someone say

[DRAFTING INSTRUCTION] *Roughly 900 characters introducing Figure 10 and the third rank it adds. The parent protocol stops at the endpoint. A participant who reads only the endpoint learns that the study measures an ISGPS grade B/C postoperative pancreatic fistula rate and is no better informed. Rank three restates the endpoint as the sentence it authorises. Cite `Bassi2017` for the fistula grading and `Conroy2018` for the adjuvant survival context.*

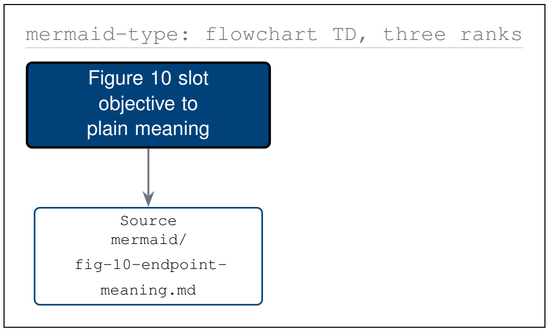

Figure 10. Each objective carried through its endpoints to the plain sentence the endpoint will authorise, with the three sentences that patients in the surveyed literature ranked highest, clear margins, being alive, and no return, marked apart.

[DRAFTING INSTRUCTION] *Replace the Figure 10 slot with the mermaid-type three-rank flowchart from* `mermaid/fig-10-endpoint-meaning.md`*. Ranks at $y = 0, -3.0, -6.4$; rank 2 text width 26 mm and rank 3 text width 30 mm; fan the objective-to-endpoint edges with looseness 0.7 rather than drawing straight diagonals through sibling boxes.*

4.3 The endpoint table, with a column the protocol does not carry

[DRAFTING INSTRUCTION] *Populate every row from* `inputs/phase-1-trial-protocol.zip`, `sections/sec-03-objectives.tex`*. The final column is the contribution of this paper and must be written for the participant, not for the statistician. Verify the widths sum to the body measure.*

Class	Endpoint	Timepoint	What it means for you
Primary	Dose-limiting toxicity rate, DL1 to DL3	Day 30	The dose you were given was tolerated at your level
Primary	Device- and procedure-related serious adverse event rate	Day 30	Nothing the device did caused you serious harm
Primary	Clavien-Dindo grade III or higher rate	Day 30	Your recovery needed no unplanned major intervention
Secondary	R0 resection rate	Day 90	The margins came back clear
Secondary	ISGPS grade B/C fistula rate	Day 30	Your pancreas join healed without a draining leak
Secondary	90-day mortality	Day 90	You were alive at three months
Secondary	Progression-free and overall survival	24 months	The cancer had not returned by this visit
Exploratory	Advisory concordance and override rate	Per procedure	Your surgeon disagreed with the model here, and was free to

4.4 The endpoint registry as a typed record

[DRAFTING INSTRUCTION] *Roughly 700 characters introducing Figure 11. An endpoint is a typed record with a definition, a timepoint, an analysis population, a provenance class, and, in this paper, a plain meaning. Drawing it as a schema forces every field to be filled and exposes any endpoint where one is empty.*

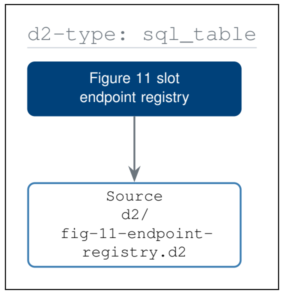

Figure 11. The endpoint registry drawn as typed records with foreign keys to the analysis populations and to the provenance classes, and with a plain-meaning row that no clinical protocol carries but that this paper treats as a required field.

[DRAFTING INSTRUCTION] *Replace the Figure 11 slot with the D2-type SQL-table figure from `d2/fig-11-endpoint-registry.d2`. Five `d2sql` nodes at the coordinates given in that file; each row a 4.4 mm ruled band; the plain-meaning row filled `pablue2` to mark it as the added field; relations drawn with crow-foot ends, the two cross-rank relations at looseness 0.85.*

4.5 What a positive result would and would not establish

[DRAFTING INSTRUCTION] *Roughly 900 characters, and write it as a proponent who intends to be believed. A clean safety readout at day 30 and an R0 rate consistent with the nationwide comparator would establish that the combination can proceed to a study that could test survival. It would not establish that the combination improves survival, and this paper does not claim otherwise. Cite `DutchCohort2025`, `FDARobot2022`, and `NCITrial2024`.*

5 Study Design Explained

5.1 Open-label, single-arm, and what that means for you

[DRAFTING INSTRUCTION] *Roughly 1,000 characters. Three terms, each with a consequence. Open-label means nothing about your treatment is concealed from you. Single-arm means there is no allocation that could assign you to a comparator, so the common fear of being randomised to a treatment you would not have chosen does not arise here. Dose-finding means the study is looking for the highest dose that is tolerated, not for the largest effect. Cite* `NCITrial2024` *for the general framing and* `onpremwhippl` *for the design.*

5.2 Your status in the study, and how it can change

[DRAFTING INSTRUCTION] *Roughly 900 characters introducing Figure 12 and the idea of a guard. Every transition in the figure carries a named condition that must evaluate true, and the protections live in the guards rather than in the states. Adapt from* `inputs/phase-1-trial-protocol.zip`, `sections/sec-04-design.tex`.

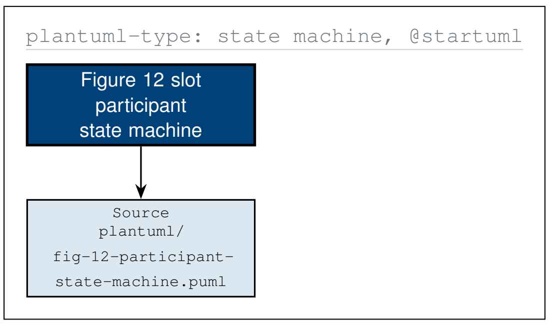

Figure 12. Participant status as a guarded state machine, screening through the 24-month endpoint, with the sentinel cohort hold drawn as an explicit state because it is the clearest evidence that the design accepts delay in exchange for safety.

[DRAFTING INSTRUCTION] *Replace the Figure 12 slot with the PlantUML-type state machine from* `plantuml/fig-12-participant-state-machine.puml`. *Single column of* `umlstate` *boxes at 2.4 cm pitch; the two off-path states at $x = 6.8$ and $x = -6.8$ aligned with the state they branch from; guards in white-filled* `umlguard` *nodes at the transition midpoint; the self-transition on* `DoseAssigned` *as a 5 mm right-hand loop at looseness 5.*

5.3 The 3+3 rule, and why you are never the untested dose

[DRAFTING INSTRUCTION] *Roughly 1,300 characters. Explain the rule operationally rather than statistically. Three participants at a dose level; if none has a dose-limiting toxicity, the next level opens; if one does, the level expands to six rather than escalating; if two or more do, escalation stops and the level below becomes the recommended dose. Add the sentinel stagger: the second participant at a new level is not dosed until the first participant's safety window has closed. State that this costs the study time and is not there for the study's benefit.*

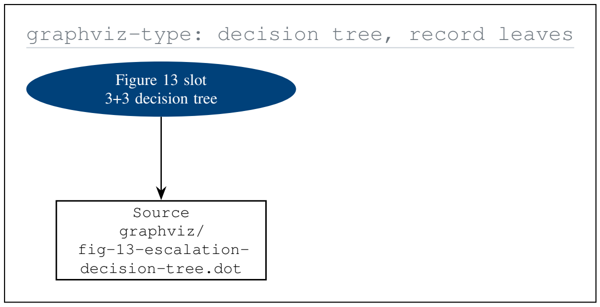

Figure 13. The 3+3 dose-escalation rule drawn as a decision tree with exactly one path from the root to each leaf, including the two stopping leaves and the study pause that convenes the Data and Safety Monitoring Board and is reported to you.

[DRAFTING INSTRUCTION] *Replace the Figure 13 slot with the graphviz-type decision tree from `graphviz/fig-13-escalation-decision-tree.dot`. Four ranks at $y = 0, -3.0, -6.4, -9.8$; branch labels placed on the upper third of each edge, not the midpoint, so sibling labels cannot collide; the sentinel-wait annotation as a `gvcluster` dashed box behind rank 1; record leaves built from stacked `gvcell` rows.*

5.4 The dose levels

[DRAFTING INSTRUCTION] *Populate the dose levels and the cohort sizes from `inputs/phase-1-trial-protocol.zip`, `sections/sec-04-design.tex` and `sections/sec-06-intervention.tex`. Keep the final column patient-facing.*

Level	Daraxonrasib dose	Cohort size	What has to be true before this level opens
DL1	Starting dose level	3, expandable to 6	Phase 0 gate signed, USL at least 7.0, IND in effect, IDE approved
DL2	One step above DL1	3, expandable to 6	DL1 cleared: 0 of 3, or 1 of 6, dose-limiting toxicities
DL3	Two steps above DL1	3, expandable to 6	DL2 cleared on the same rule, and DSMB review at the cohort boundary

Planned enrollment is up to 18 participants across the three levels. Enrollment is staggered: no participant enters a level while a sentinel window at that level remains open.

5.5 The Phase 0 gate that runs before any patient

[DRAFTING INSTRUCTION] *Roughly 1,100 characters. At least 1000 simulated procedures across at least two independent simulation frameworks, reproducing motion within a 2 mm positional and 0.5 N force tolerance, with a documented Unified Safety Level of at least 7.0. Cite `asmevv40` and `nasastd7009` for the credibility standards this is built against, `cfr312paiuni` for the Subpart J overlay, and `pdac060s2030` for the simulation source. State clearly that a simulated worst case is not a human worst case.*

5.6 Stopping rules, stated before they are needed

[DRAFTING INSTRUCTION] *Roughly 900 characters listing the prespecified pause and stopping triggers from `inputs/phase-1-trial-protocol.zip`, `sections/sec-04-design.tex` §4.6. Include the device-specific triggers, because these are the ones a participant worried about the robot will look for: any breach of a force cap, any entry into a vascular no-fly corridor, any emergency-stop latency above budget, and any loss of the heartbeat bus.*

6 Who Can Join, and Who Decides

6.1 Eligibility as a two-way gate

[DRAFTING INSTRUCTION] *Roughly 1,100 characters. Standard eligibility runs in one direction: the protocol tests the participant and returns eligible or screen failure. Under the author's proposed legislation the participant is also doing the selecting, and the second direction leaves a record. Cite* `hr9510billv5` *and* `paibillprior`. *Adapt the criteria from* `inputs/phase-1-trial-protocol.zip`, `sections/sec-05-population.tex` *§5.1 and §5.2.*

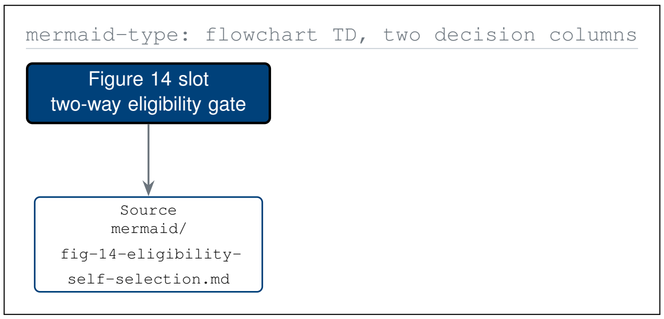

Figure 14. Eligibility drawn as two decision columns running against each other, what the protocol asks of the participant and what the participant may ask of the protocol, converging on a single enrollment node that requires both to be satisfied.

[DRAFTING INSTRUCTION] *Replace the Figure 14 slot with the mermaid-type two-column flowchart from* `mermaid/fig-14-eligibility-self-selection.md`. *Two* `mmlane` *boxes at $x = -3.4$ and $x = 4.6$; four* `mmdec` *diamonds per lane at 2.4 cm pitch; the two edges out of* `START` *at looseness 0.75 so neither clips a lane title; the two "all yes" edges into enrollment at looseness 0.8.*

6.2 Inclusion and exclusion, in plain terms

[DRAFTING INSTRUCTION] *Populate both tables from* `inputs/phase-1-trial-protocol.zip`, `sections/sec-05-population.tex` *§5.1 and §5.2. Keep the second column in the participant's own terms: an inclusion criterion is a statement about them, not about the study.*

You may join if	Why this criterion exists
You are 18 or older	The study has no paediatric safety data and does not enrol children
Your tumour is confirmed PDAC on tissue	The drug and the operation are both specific to this disease
Your tumour carries a KRAS G12 mutation	Daraxonrasib acts on RAS(ON); without the mutation there is no rationale
Your tumour is resectable or borderline resectable	The operation must be able to achieve a clear margin
Your performance status is ECOG 0 or 1	A Whipple is a major operation and the study will not expose a frailer participant to it
Your organ function is adequate	Both the drug and the operation load the liver, kidneys, and marrow

You may not join if	Why this criterion exists
Your disease is metastatic	A curative-intent resection is not the right operation
You have had prior pancreatic resection	The anatomy the platform is validated against is not present
You have an uncontrolled intercurrent illness	The perioperative risk cannot be attributed cleanly
You are pregnant or breastfeeding	The drug has no safety data in pregnancy
You cannot attend the follow-up schedule	The safety endpoints depend on the day-30 and day-90 visits

[DRAFTING INSTRUCTION] *Add the remaining criteria from §5.1 and §5.2 of the parent protocol so both tables are complete, and verify each table lands exactly on the body measure with the 5.4cm plus Y split.*

6.3 Was this tested on people like me?

[DRAFTING INSTRUCTION] *Roughly 1,000 characters answering ChatGPT concern 12 honestly. The honest answer for a first-in-human study of at most eighteen participants is that the trial population is too small to establish subgroup performance, and that what the protocol can commit to is reporting the composition. Cite* `FDADiver2017` *for the age, race, and ethnicity reporting expectation and* `FDARealW2025` *for real-world performance monitoring after the study.*

Characteristic	Reported	What the study can and cannot say
Age, sex, race, ethnicity	Yes, per participant	Composition is reported; subgroup performance is not estimable at $n \leq 18$
Body mass index	Yes, per participant	Relevant to port placement and to force at the tip
Prior therapy	Yes, per participant	Prior radiation changes tissue planes and is recorded
Tumour location and vessel contact	Yes, per participant	Determines how much of the no-fly corridor is in play
Comorbidity burden	Yes, per participant	Feeds the Clavien-Dindo interpretation at day 30

6.4 Booking a slot yourself, and getting an answer from a person

[DRAFTING INSTRUCTION] *Roughly 1,000 characters introducing Figure 15. Under current practice a participant learns a trial exists from a clinician who happens to know about it. H. R. 9510 v5 proposes a direct channel with a stated service level. State the target: five business days from first request to a confirmed screening visit, with a 72-hour provisional hold so nobody is rushed. State the protection: every branch notifies the treating oncologist, because the channel is additional and not a replacement. Cite* `hr9510billv5` *and* `paiautospons`*.*

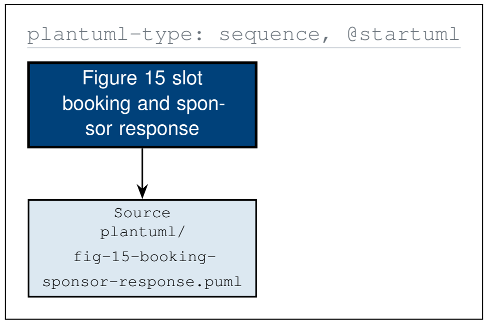

Figure 15. Booking a screening slot directly and receiving an answer from a named human at the sponsor, with activation bars showing who is blocked at each step and every branch of the eligibility fragment notifying the participant's own oncologist.

[DRAFTING INSTRUCTION] *Replace the Figure 15 slot with the PlantUML-type sequence from* `plantuml/fig-15-booking-sponsor-response.puml`*. Five lifelines at* $x = 0, 3.8, 7.6, 11.4, 15.2$*;* `umlact` *activation bars spanning exactly the messages they enclose; message labels above the arrow, never on it; the single self-message as a 4 mm loop at looseness 6; the* `alt` *fragment as a 0.5pt* `protogray` *rectangle with a corner tab and a dashed divider.*

6.5 The five choices you keep

[DRAFTING INSTRUCTION] *Roughly 800 characters introducing Figure 16. A consent form presents every choice at once at the moment of greatest stress. The same five choices, revealed as layers that accumulate rather than replace, are each live at a different point in the study, and not one of them is exercised on the day of surgery. Cite* `NCIConsent24` *and* `cfr050paiuni`*.*

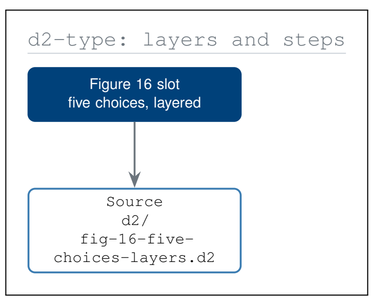

Figure 16. The five choices the participant keeps, revealed as accumulating layers with the choice, the interval in which it is live, and what happens if the answer is no, so that no decision that matters is taken at the moment of least capacity.

[DRAFTING INSTRUCTION] *Replace the Figure 16 slot with the D2-type layered figure from* `d2/fig-16-five-choices-layers.d2`*. Five* `d2step` *panels stacked at 3.6 cm pitch; earlier panels redrawn as* `d2ghost` *0.4pt outlines with no text so the reveal reads as accumulation; the panel-to-panel connector on the left margin at looseness 0.7.*

7 What Happens in the Operating Room

7.1 The room, the cabinet, and the cable that is not connected

[DRAFTING INSTRUCTION] *Roughly 1,200 characters introducing Figure 17. State the three things that cross the network boundary before a procedure, a signed model build, a protocol version, and the pinned registry entry; state the two that cross after it, reports to the FDA and the IRB; and state that nothing crosses during it. Cite `FDACyber2026` for the cybersecurity expectation and `onpremwhippl` for the on-premises design. Note that the stack topology is adapted from the author's NSCLC single-patient journey [6] and re-scoped to PDAC: eight arms rather than four, three anastomoses rather than one bronchial closure, and a vascular exclusion envelope the thoracic configuration has no equivalent of.*

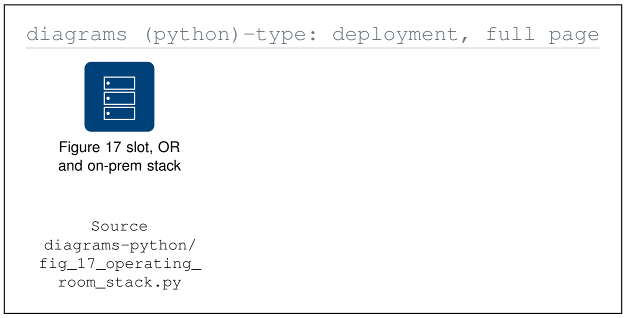

Figure 17. The operating room and the on-premises stack drawn as a deployment architecture, with the sterile field, the non-sterile side, the cabinet, the evidence store, and the network boundary that carries no intra-procedural route at all.

[DRAFTING INSTRUCTION] *Replace the Figure 17 slot with the full-page Diagrams (Python)-type deployment from `diagrams-python/fig_17_operating_room_stack.py`. Five `dgcluster` containers in the two-column arrangement given in that file; every tile a `\dgnode` with the named pictogram; the absent cable interrupted by a `\pxmark` with its label in a white-filled node above the cross.*

7.2 The eight arms, and what limits them

[DRAFTING INSTRUCTION] *Roughly 1,300 characters. Per-arm tip force at most 3 N, cumulative across arms at most 18 N, positional tolerance 2 mm, force reproducibility 0.5 N, a 10 kHz heartbeat broadcast bus, and a vascular no-fly corridor 4 mm either side of the superior mesenteric and portal veins. State the worst observed value from the Phase 0 simulation set beside each cap, and state that a simulated worst case is not a human worst case. Cite `pdac060s2030` and `iec8060127`.*

7.3 Who is controlling the operation

[DRAFTING INSTRUCTION] *Roughly 1,400 characters, and treat this as the most important subsection in the paper. This is the single most frequently raised concern in every surveyed source [7, 5, 4]. The model proposes; a deterministic gate tests the proposal against the force caps and the no-fly corridor; the surgeon approves, modifies, or declines, and declining requires no justification; the signed command reaches the arms. There is no message from the model to the arms. Cite `LeeAuto2024X` for the published finding that no FDA-cleared surgical robot exceeds task-level autonomy under supervision, and `FDARobot2022`.*

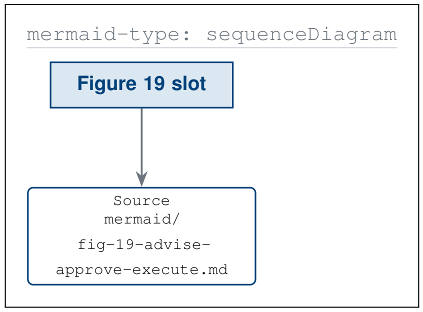

Figure 19. One operative step message by message across five lifelines, with the deterministic gate between advice and approval, and no arrow at all from the model to the arms, which is the safety argument this figure makes by an absence.

[DRAFTING INSTRUCTION] *Replace the Figure 19 slot with the mermaid-type sequence from* `mermaid/fig-19-advise-approve-execute.md`*. Five* `mmactor` *headers at 3.6 cm pitch;* `mmlife` *lifelines from* -0.6 *to* -12.6*;* `mmact` *activation bars; self-messages as 4 mm right-hand loops at looseness 6; the* `alt` *fragment with a corner tab; the two closing notes at* $y = -11.2$ *and* -12.2 *so they do not collide with the last message. Figure 18 precedes this figure in reading order and is placed above it.*

7.4 How fast stopping actually is

[DRAFTING INSTRUCTION] *Roughly 1,100 characters introducing Figure 18. Three budgets: a 0.1 ms heartbeat period, a 3 ms cross-arm halt, and a 500 ms system-wide guarantee. Give the human referent: a blink is roughly 100 to 150 ms, so every arm is held before a blink completes. Name the three parties who may press stop: the operating surgeon, the second operator station, and the independent safety monitor. Cite* `ieee7009`*.*

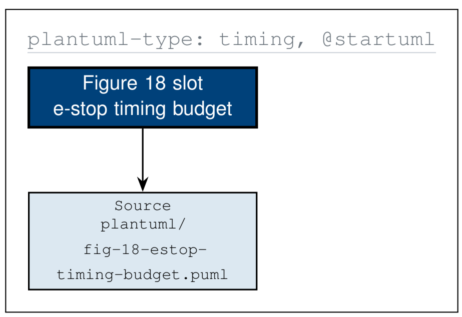

Figure 18. The emergency-stop chain on a logarithmic millisecond axis, from the instant the surgeon presses stop through the bus broadcast and arm deceleration to zero tip force, with all three protocol budget lines marked across every timeline.

[DRAFTING INSTRUCTION] *Replace the Figure 18 slot with the PlantUML-type timing diagram from* `plantuml/fig-18-estop-timing-budget.puml`*. Five stacked 14 cm bands at 1.6 cm pitch; a logarithmic axis from 0.1 to 1000 ms with decade ticks; state segments as 4 mm rectangles filled* `protoblue` *while asserted,* `pablue2` *while armed,* `pagraym` *while idle; budget markers as 0.8pt dashed verticals spanning all five bands with the label in a white-filled node. Place this figure before Figure 19 on the page.*

7.5 The vascular no-fly corridor and the force envelope

[DRAFTING INSTRUCTION] *Roughly 900 characters introducing Figure 21. A force cap does not prevent an arm entering a vascular corridor, and a corridor does not cap the force applied inside permitted tissue. The pairing is the safety property, and the figure draws them as sibling containers inside one envelope for that reason.*

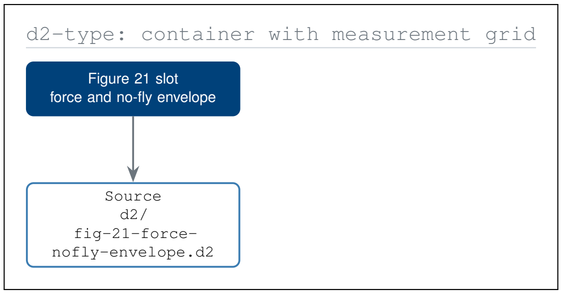

Figure 21. The operative envelope drawn as two sibling containers, the force limits with their observed headroom on the left and the vascular no-fly geometry on the right, joined by the deterministic gate that rejects any plan violating either.

[DRAFTING INSTRUCTION] *Replace the Figure 21 slot with the D2-type envelope figure from* `d2/fig-21-force-nofly-envelope.d2`*. One outer* `d2cont`*, two* `d2cont2` *sub-containers 1.6 cm apart; the force sub-container as a 5-row measurement grid using* `\hbarrow` *with the cap as a dashed vertical and the observed value as a filled bar; the geometry sub-container as a schematic with two 1.4 mm vessel bands, a hatched* `pagraym` *corridor, and eight arm-tip discs, none inside the corridor; the invariant strip in* `d2dark` *along the foot.*

7.6 What could fail, and what catches it

[DRAFTING INSTRUCTION] *Roughly 1,200 characters introducing Figure 20. Four intermediate events must all occur for harm to reach the participant, which is what the AND gate at the apex means. Name the four barriers, and name the one gap: registration drift beyond 2 mm is caught by the deterministic gate alone. Do not omit the gap.*

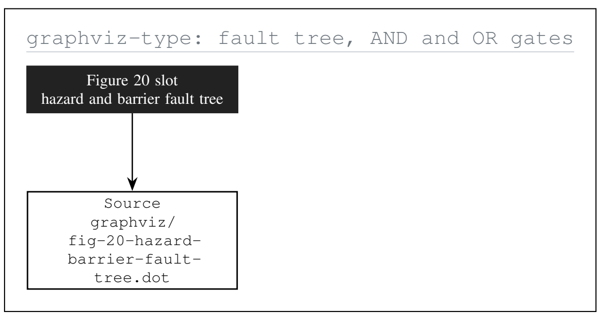

Figure 20. The hazard fault tree with the top event behind an AND gate, so that harm requires all four intermediate events together, and with the single-barrier gap at registration drift marked in medium-dark gray rather than quietly omitted.

[DRAFTING INSTRUCTION] *Replace the Figure 20 slot with the graphviz-type fault tree from* `graphviz/fig-20-hazard-barrier-fault-tree.dot`*. Use* `\umlgateand` *and* `\umlgateor` *for the five gates; ten basic events as 9 mm* `gvcircle` *nodes on the lowest rank; barrier annotations in the right margin attached by 0.5pt dashed leaders so they never sit between two events; the two cross-branch edges at looseness 0.9.*

7.7 Conversion to open surgery

[DRAFTING INSTRUCTION] *Roughly 800 characters. Conversion is a recorded outcome, not a failure of the study or of the participant, it does not end participation, and the decision belongs to the operating surgeon alone. Give the comparator conversion rate context from* `DutchCohort2025` *and state that backup instruments and an open tray are in the room for every procedure.*

8 Stopping, Withdrawing, and Changing Your Mind

8.1 Four routes out, and what each does to your record

[DRAFTING INSTRUCTION] *Roughly 1,200 characters introducing Figure 22. Participants are routinely told they may withdraw at any time and are almost never told what withdrawal does to the data already collected. The honest answer depends on which route out is taken, and there are four: stop the drug only; stop study procedures but keep follow-up; full withdrawal; and withdrawal of consent for data use as well. Adapt from* `inputs/phase-1-trial-protocol.zip`, `sections/sec-07-discontinuation.tex`, *and cite* `ecfr2026part50` *and* `NCIConsent24`.

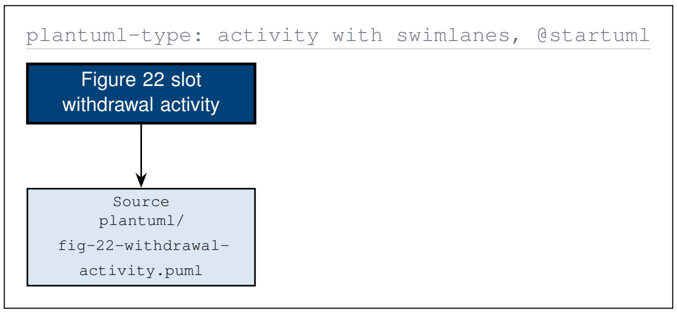

Figure 22. The four routes out of the study drawn as a swimlaned activity, with one lane for the participant, one for the site team, and one for the data and specimens, so each route's consequence for the record is visible beside the route itself.

[DRAFTING INSTRUCTION] *Replace the Figure 22 slot with the PlantUML-type activity from* `plantuml/fig-22-withdrawal-activity.puml`. *Three vertical* `umlpkg` *swimlanes at $x = -5.0, 0.6, 6.6$ sharing one top and one bottom edge; actions as* `umlstate` *boxes at 2.2 cm pitch; one* `mmdec` *decision at $(0.6, -3.0)$; fork and join as* `umlbar` *rectangles; four* `umlfinal` *terminals on one rank; cross-lane edges with right-angle routing, the two that change rank at looseness 0.85.*

Route	What stops	What happens to data and specimens
A	Daraxonrasib only	Everything already collected stays in the safety analysis; new data continues to be collected
B	Study procedures, follow-up continues	Existing data retained; new data limited to survival status; specimens retained unless you ask for destruction
C	All study participation	Data already collected cannot be removed from the safety analysis; nothing new is collected; specimens destroyed on request
D	Participation and future data use	Identifiable data segregated from new analysis; aggregate safety data already filed with the FDA cannot be recalled

[DRAFTING INSTRUCTION] *Do not soften routes C and D. Both statements are true and unwelcome, and omitting them would make routes A and B less believable.*

8.2 Consent as a state, not a signature

[DRAFTING INSTRUCTION] *Roughly 1,100 characters introducing Figure 23. A consent form implies a one-time act. In a Physical AI trial the thing consented to can change while the participant is still enrolled, because an AI-enabled device can be updated. This protocol freezes the software version at consent and forces a re-consent on any change, so the participant is never operated on by software they have not been told about. Cite* `FDATrans2024` *and* `FDARealW2025`.

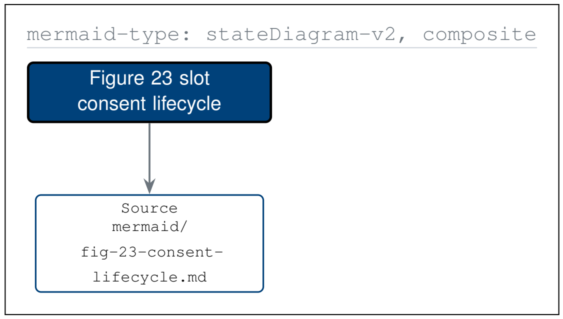

Figure 23. Consent drawn as a state machine with a nested enrolled composite, withdrawal reachable from every state, and a re-consent transition the participant did not cause, triggered by a software version change, a new risk, or a protocol amendment.

[DRAFTING INSTRUCTION] *Replace the Figure 23 slot with the mermaid-type state diagram from* `mermaid/fig-23-consent-lifecycle.md`*.* `umlinit` *at* $(0, 1.4)$*, two* `umlfinal` *discs at* $y = -13.0$*; the enrolled composite as a* `umlpkg` *fitted over five* `umlstate` *substates at 1.4 cm pitch; transitions out of the composite leave the composite border rather than a substate, at looseness 0.9; the two notes as* `umlnote` *boxes with 0.5pt dashed leaders.*

8.3 Discontinuation the study may initiate

[DRAFTING INSTRUCTION] *Roughly 900 characters. Distinguish clearly between the participant stopping and the study stopping the participant. Populate the triggers from* `inputs/phase-1-trial-protocol.zip`*,* `sections/sec-07-discontinuation.tex` *§7.2, and state that in every case the participant is told the reason, in writing, and returns to standard care.*

Trigger	What follows for you
Dose-limiting toxicity at your level	Drug held or discontinued; surgical follow-up and every safety assessment continue
Disease progression before surgery	Curative-intent resection is no longer the right operation; you are referred back for systemic therapy
Intercurrent illness precluding surgery	Participation ends; standard-of-care pathway resumes without penalty
Study-wide pause or halt	You are told, in writing, within the timeframe the protocol specifies; follow-up continues
Withdrawal of the IND or the IDE	Participation ends; the sponsor's obligation for research-related injury care continues

8.4 Lost to follow-up

[DRAFTING INSTRUCTION] *Roughly 600 characters. Define what counts as lost to follow-up, state the contact attempts the protocol requires before that determination, and state that a participant who reappears is resumed rather than excluded. Source from §7.3 of the parent protocol.*

9 Your Visits, Your Data, Your Robot Instructions

9.1 The calendar, and what it costs you

[DRAFTING INSTRUCTION] *Roughly 1,000 characters introducing Figure 25. Convert the Schedule of Activities into a calendar with durations and then total it: about 18 visits, 8 to 10 inpatient days, roughly 27 days of contact over 24 months. State that the intervals, not the illustrative calendar dates, are what the protocol fixes. Cite* `NCICosts2024` *for the practical burden framing.*

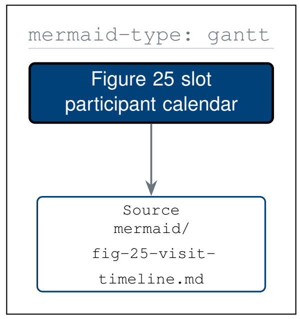

Figure 25. The participant calendar from screening through the 24-month endpoint, drawn with real durations rather than crosses in a matrix, and with the four critical milestones, surgery, day 30, day 90, and the endpoint, marked to the axis.

[DRAFTING INSTRUCTION] *Replace the Figure 25 slot with the mermaid-type Gantt from* `mermaid/fig-25-visit-timeline.md`*. A 15 cm time axis spanning September 2026 to September 2028; five alternating section bands drawn before any bar; 3.6 mm bars with 2.2 mm clear space between rows; task labels anchored east in a 4.6 cm left gutter, never inside a bar; the four critical milestones with a 0.5pt drop line to the axis.*

Phase	Visits	Inpatient days	Clinic days	Notes for planning
Screening	4	0	about 4	Imaging and tissue work; no study procedure before consent
Baseline and surgery	2	8 to 10	1	The single inpatient stay of the study
Early recovery	3	0	3	The restart advisory is a conversation, not a procedure
Confirmatory	2	0	2	Day 90 carries the two results most people want
Long-term	7	0	7	One half-day every 12 weeks for 24 months
Total	**18**	**8 to 10**	**about 17**	Roughly 27 days of contact over 24 months

9.2 Everything recorded about you

[DRAFTING INSTRUCTION] *Roughly 1,300 characters introducing Figure 24. List every data type: arm motion, tip force, vessel proximity across 640 sensor channels; operative video and endoscope feed; vitals, laboratory values, CA 19-9, drain output; the case report form. State retention in years per store, and state the read list per store. Cite* `ecfr2026part11` *for the electronic-records standard and* `FDATrans2024` *for the transparency principles.*

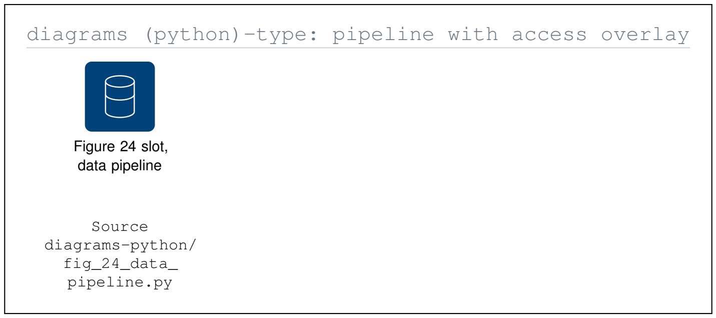

Figure 24. Your data through four stages, capture, write-once hash chain, the identified and de-identified stores, and what leaves, with the read list attached to every store and no edge at all for a principal that may not read it.

[DRAFTING INSTRUCTION] *Replace the Figure 24 slot with the Diagrams (Python)-type pipeline from `diagrams-python/fig_24_data_pipeline.py`. Four stage clusters plus one access-control container; retention printed on a third label line beneath each store tile so no leader is needed; a principal that cannot read a store is joined by nothing at all, and the figure's foot note says that the absence is the access statement.*

9.3 Who can read what, and for how long

[DRAFTING INSTRUCTION] *Populate this table from the `READ_ACCESS` and `RETENTION_DAYS` dictionaries in `diagrams-python/fig_24_data_pipeline.py`, converting days to years.*

Store	Retention	Who may read it
Raw telemetry	10 years	Site PI, Sponsor-Investigator, and you on request
Operative video	10 years	Operating surgeon, Site PI, and you on request
Case report form	15 years	Site PI, CRO monitor, and you on request
Hash chain	15 years	Sponsor-Investigator, FDA on inspection, and you on request
Identified store	15 years	Site PI, and you on request
De-identified store	10 years	Sponsor-Investigator, DSMB, study statistician
Analysis outputs	15 years	Everyone above, and the public at publication

Your data is not used to train a commercial model without a separate consent. That is a statement about what does not happen, and Figure 24 draws it as an absent edge rather than asserting it in a caption.

9.4 Cybersecurity, in the terms the question is asked

[DRAFTING INSTRUCTION] *Roughly 900 characters answering ChatGPT concern 11 directly. The question is "could someone hack the system", and the answer is architectural rather than reassuring: during a procedure there is no network route out of the building, so the attack surface is the building. State what crosses before and after, and state what remains true if hospital connectivity fails entirely, which is that the procedure is unaffected. Cite `FDACyber2026`.*

9.5 The robots you will meet, and the ones you will not

[DRAFTING INSTRUCTION] *Roughly 1,100 characters introducing Figure 26. The source instruction set covers ten robot types [8]. In this protocol two are certain, four are possible depending on recovery, and four are not used. Knowing which is which is itself an answer to the question of what the participant is agreeing to. State the re-scoping explicitly: the source sheets are mixed-oncology and illustrated with raster images, and both are changed here.*

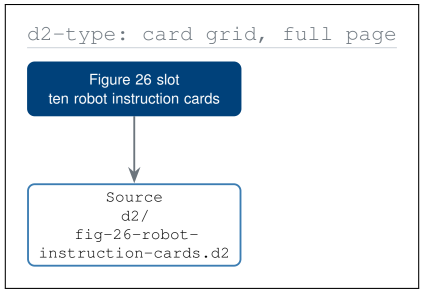

Figure 26. Ten robot-type instruction cards re-scoped from general oncology to this PDAC Whipple protocol, each with three numbered instructions and a re-scoping note, and with the four robot types this protocol does not use marked as such.

[DRAFTING INSTRUCTION] *Replace the Figure 26 slot with the full-page D2-type card grid from `d2/fig-26-robot-instruction-cards.d2`. Two columns by five rows of 7.0 cm by 3.2 cm cards with 0.6 cm of clear space in both axes; each card a `d2cont` with a `d2key` title band, exactly three numbered instruction lines, and a `d2gray` foot strip; the eight-arm surgical card drawn with a 1.2pt border and every other card at 0.8pt. No raster image.*

9.6 Software changes during your participation

[DRAFTING INSTRUCTION] *Roughly 800 characters. The software version is frozen at consent and recorded in the version registry. Any change forces a re-consent, and until the participant acts no study procedure proceeds under the changed version. Declining the change ends participation and returns the participant to standard care with no penalty. Cite `FDARealW2025` and forward to Figure 23.*

10 The Numbers Behind the Reassurance

10.1 The provenance rule this section obeys

[DRAFTING INSTRUCTION] *Roughly 900 characters. Every quantitative claim in this section carries one of four class letters, and the letter appears on the same line as the number. M means measured in a cited human series; C means a contemporaneous nationwide comparator; S means an author simulation and not human data; P means a protocol-specified limit and not a result. State that two of the nine headline values are class S and name them. Cite* `Jauniaux2025` *for why this discipline matters: expectation and experience diverge systematically in robotic surgery, and expectation is usually set by marketing.*

10.2 The reassurance dashboard

[DRAFTING INSTRUCTION] *Roughly 1,200 characters introducing Figure 27 panel by panel, and do not skip the unfavourable panels. Panel A: fistula and mortality falling with institutional volume, cite* `DutchCohort2025`*. Panel B: 8.2 percent of the 1000 Phase 0 simulated procedures halted by a safety mechanism, cite* `pdac060s2030`*. Panel C: every observed worst case inside its cap. Panel D: the survival context with the honest denominator, cite* `Siegel2025` *and* `Conroy2018`*. Panel E: how many concerns get which class of answer.*

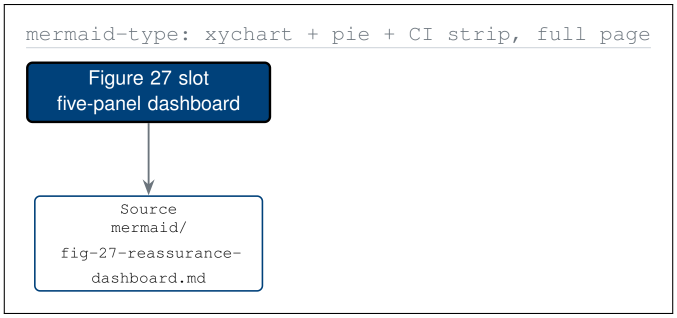

Figure 27. Five panels of quantitative reassurance on one page, learning-curve outcomes, Phase 0 simulation endings, safety-limit headroom, the survival context with its honest denominator, and the class of answer each documented concern receives.

[DRAFTING INSTRUCTION] *Replace the Figure 27 slot with the full-page mermaid-type dashboard from* `mermaid/fig-27-reassurance-dashboard.md`*. Five stacked panels inside one frame separated by 0.4pt* `pagraym` *hairlines; Panel A with* `\vbarcol` *plus a 0.8pt polyline; Panel B with* `\donutseg` *and leader lines to labels outside the ring; Panel C with* `\vbarpair`*; Panels D and E as* `tabularx` *tables at the panel measure.*

10.3 Robotic pancreatoduodenectomy, by learning-curve phase

[DRAFTING INSTRUCTION] *Populate from* `DutchCohort2025`*. Add the class letter to every numeric column heading. State the protocol's response in the final column: enrollment is restricted to a site already past phase 3, so the learning curve is not paid for by the participant.*

Learning-curve phase	ISGPS B/C fistula (C)	90-day mortality (C)	What this protocol does about it
Phase 1, cases 1 to 80	24.0 percent	8.0 percent	Not eligible to enrol; the site must be past this phase
Phase 2, cases 81 to 180	18.5 percent	6.0 percent	Not eligible to enrol
Phase 3, cases 181 to 400	14.0 percent	4.5 percent	Minimum eligible experience for a participating site
Phase 4, cases 401 and above	11.0 percent	3.5 percent	Preferred; operator case counts are published to participants

10.4 Safety-limit headroom

[DRAFTING INSTRUCTION] *Populate from* `onpremwhippl` *for the caps and* `pdac060s2030` *for the observed values. Mark the cap column P and the observed column S, and repeat in the caption that a simulated worst case is not a human worst case.*

Limit	Protocol cap (P)	Worst observed (S)	Headroom
Tip force, any single arm	3.0 N	2.1 N	30 percent
Tip force, all arms summed	18.0 N	12.4 N	31 percent
Cross-arm emergency stop	3.0 ms	1.7 ms	43 percent
System-wide emergency stop	500 ms	310 ms	38 percent
Positional error	2.0 mm	1.3 mm	35 percent
Force reproducibility	0.5 N	0.31 N	38 percent

10.5 The survival context, with the honest denominator

[DRAFTING INSTRUCTION] *Roughly 1,000 characters and the most important honest passage in the paper. Pancreatic cancer five-year overall survival across all stages is 13 percent [9]. Resected PDAC after adjuvant FOLFIRINOX is materially better [10]. An R0 margin is materially better than R1. A Phase 1 study of at most eighteen participants cannot move a survival curve and this paper does not claim it will. What it can establish is whether the combination is safe enough to be tested in a study that could.*

10.6 Where the numbers come from

[DRAFTING INSTRUCTION] *Roughly 800 characters introducing Figure 28. Nine quoted values, six sources, four evidence classes, and one rule: a value with no incoming edge does not appear in this paper.*

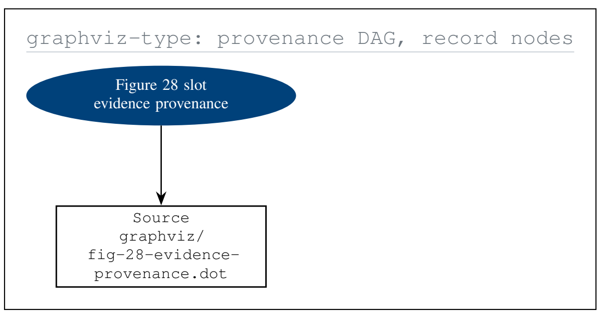

Figure 28. Every quantitative claim in this paper traced through its source to its class of evidence, with the two class-S values, which are simulated rather than measured, marked by dashed edges so a reader can separate them at a glance.

[DRAFTING INSTRUCTION] *Replace the Figure 28 slot with the graphviz-type provenance DAG from* `graphviz/fig-28-evidence-provenance.dot`*. Three ranks at $x = 0, 6.4, 12.4$; nine* `gvrec` *record nodes on rank 1 at 1.9 cm pitch; six* `gvbox` *sources on rank 2; four* `gvnode` *class ellipses on rank 3; class-S edges dashed* `pagrayd`*; where three or more values share a source, fan the outer edges at looseness 0.8.*

10.7 What would change the conclusion

[DRAFTING INSTRUCTION] *Roughly 700 characters. Name the findings that would make this paper wrong: a device-related serious adverse event rate above the prespecified halt threshold, an R0 rate below the comparator range, a fistula rate above phase-1 learning-curve levels at an experienced site, or any emergency-stop latency measured above budget in a human procedure. State that each has a prespecified stopping rule attached, and forward to §5.6.*

11 Accountability, Oversight, and Who Answers

11.1 The question, stated as patients state it

[DRAFTING INSTRUCTION] *Roughly 1,000 characters. The concern in the surveyed literature is not that nobody is responsible; it is that everybody is, which amounts to the same thing [4, 7]. When a human surgeon errs the chain of liability is clear. When a semi-autonomous movement or a model-generated suggestion contributes to harm, the participant is told that the surgeon, the hospital, the developer, and the manufacturer are all involved, and learns nothing. Answer with the one-accountable-party rule and forward to Figure 29.*

11.2 One accountable party per failure class

[DRAFTING INSTRUCTION] *Roughly 1,100 characters introducing Figure 29 and its invariant: exactly one accountable party per row, verified across nine rows. Explain why the invariant matters: two accountable parties in one cell is the same as none. Populate the oversight bodies from* `inputs/phase-1-trial-protocol.zip`, `sections/sec-10-oversight.tex`, *and cite* `paisitedocpk`.

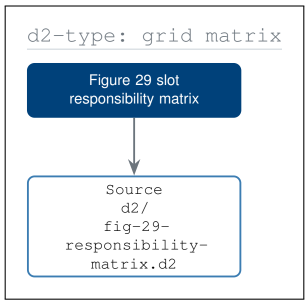

Figure 29. The responsibility matrix with exactly one accountable party in every row and a final column, the only black fill in the figure, naming the person the participant should actually call when that particular thing goes wrong.

[DRAFTING INSTRUCTION] *Replace the Figure 29 slot with the D2-type matrix from* `d2/fig-29-responsibility-matrix.d2`. *Seven columns by ten rows of* `d2cell` *rectangles sharing rules; header row* `d2cellh`; *label column* `d2celll`; *an A cell* `protoblue` *with white bold text, R* `pablue1`, *C* `pablue2`, *I* `pagray1`; *the "you call" column in* `padark` *with white text; a four-key legend strip beneath.*

If this goes wrong	Accountable party	Who you call
A motion during your operation	Operating surgeon	The surgeon, by name, recorded in your consent
A device malfunction or a halt	Sponsor-Investigator	The Sponsor-Investigator
A drug dose or a restart decision	Sponsor-Investigator	The Site PI first, then the Sponsor
A serious adverse event	Sponsor-Investigator	The Site PI, on the 24-hour line
A decision to pause the study	DSMB	You do not need to call; you are told
Your data being mishandled	Site PI	The Site PI, then the IRB directly
A software version change	Sponsor-Investigator	You are re-consented before anything proceeds
A bill you did not expect	Sponsor-Investigator	The study coordinator, then the Sponsor
A complaint about how you were treated	IRB	The IRB directly, without going through the site

11.3 The oversight bodies, and what each may actually do

[DRAFTING INSTRUCTION] *Roughly 1,200 characters. Distinguish authority from advice. The DSMB may halt the study; the Physical AI Safety Review Committee may reassess the Unified Safety Level and require a re-gate; the IRB may suspend approval and receives complaints directly from participants; the CRO monitor may raise a finding but cannot stop a procedure; the independent safety monitor in the room may stop a procedure and cannot change the protocol. Source from* `sections/sec-10-oversight.tex` *of the parent protocol.*

Body	Independent of the sponsor	Strongest action it may take
Data and Safety Monitoring Board	Yes	Halt the study at a cohort boundary or immediately
Physical AI Safety Review Committee	Yes	Require re-gating and a Unified Safety Level reassessment
Institutional Review Board	Yes	Suspend or withdraw approval; receives your complaints directly
Contract Research Organisation monitor	Contracted by the sponsor	Raise a finding to closure; cannot stop a procedure
Independent safety monitor, in the room	Yes, per procedure	Stop the procedure; cannot change the protocol
Operating surgeon	No	Stop, convert, or decline any model-proposed motion

11.4 Team experience, disclosed rather than implied

[DRAFTING INSTRUCTION] *Roughly 900 characters answering ChatGPT concern 5 directly. The concern is not general surgical experience; it is experience with this exact platform, this software version, this autonomy level, and this operation. State what the protocol publishes to the participant: the operating surgeon's robotic pancreatoduodenectomy case count, the site's position on the learning curve, the proctoring arrangement, and whether trainees participate. Cite* `DutchCohort2025` *for why case count is the right question.*

11.5 The audit trail, and your right to read it

[DRAFTING INSTRUCTION] *Roughly 1,000 characters. Every model advisory, every gate decision, every surgeon approval, and every safety-limit event is written to a hash-chained record bound to a deterministic seed and a repository commit, under 21 CFR part 11 [11]. The participant may request their own extract. State plainly what that extract will and will not contain, and state that the chain cannot be edited after the fact, which is the property that makes the record worth requesting. Cite* `cfr312paiuni`*.*

11.6 What happens if a rule in this section is broken

[DRAFTING INSTRUCTION] *Roughly 700 characters. Name the escalation route for each of the three most likely breaches: an unreported adverse event, an unauthorised data access, and a procedure performed under a software version the participant did not consent to. Each route ends at a body independent of the sponsor, and each is available to the participant without going through the site.*

12 Patient Rights, Costs, and H. R. 9510 v5

12.1 The premise, stated as legislation would state it

[DRAFTING INSTRUCTION] *Roughly 1,200 characters. The cancer patient, not the doctor, the nurse, the trial sponsor, the Institutional Review Board, or the regulator, is the priority participant in a United States oncology clinical trial. That premise is the subject of the author's proposed legislation, cited throughout this paper as H. R. 9510 v5 [2], which supersedes the earlier H. R. 9501 to H. R. 9507 numbering in* `inputs/patient-priority-physical-ai.zip` *[3]. Cite* `congress9510` *and* `fedlawhr9510` *for the enactment argument.*

What the bill would establish	What this protocol already does voluntarily
Participant self-selection of trials, with 24/7 booking	Direct booking channel with a 72-hour provisional hold and a five-business-day target, Figure 15
Participant choice of robot and autonomy level	Autonomy level set per operative step and recorded in the consent, Figure 16
Participant procedural modification authority	Any excluded step is performed manually and the exclusion is logged and reported
Verification before generation, with published gates	Phase 0 gate at 1000 simulations across two frameworks, USL at least 7.0, Figure 13
Real-time participant to sponsor communication	Named human at the sponsor answers within one business day, logged verbatim
Participant health data self-custody	Per-store read lists published; your own copy available at any time, Figure 24

[DRAFTING INSTRUCTION] *Verify every row in this table against* `inputs/patient-priority-physical-ai.zip` *and restate each bill provision in the numbering of H. R. 9510 v5 rather than the superseded H. R. 9501 to 9507 range.*

12.2 Your rights during the study, enumerated

[DRAFTING INSTRUCTION] *Roughly 1,300 characters. Enumerate the rights that the regulations already give, and mark which ones this protocol extends beyond the minimum. Ground each in 21 CFR part 50 [12] and the NCI's own consent guidance [13], and mark the extensions clearly, because a paper that presented a legal minimum as a favour would deserve the distrust the surveyed literature reports.*

12.3 What this costs you

[DRAFTING INSTRUCTION] *Roughly 1,300 characters introducing Figure 30 and this table. Be specific and do not round the uncomfortable parts away. The sponsor pays the drug, the platform, study-only imaging and laboratory work, protocol-required visits, research-related injury care, and travel to a stated cap. Your usual arrangement covers standard surgical care and the inpatient stay exactly as it would outside the study. Two items are not settled by this protocol: lost income, and continued drug access after study closure. Cite* `NCICosts2024`.

Item	Who pays	Note
Daraxonrasib	Sponsor	Supplied for the whole study period
Robotic platform use	Sponsor	No charge reaches you
Study-only imaging and laboratory work	Sponsor	Anything the protocol adds beyond routine care
Protocol-required visits	Sponsor	The 18 visits of Figure 25
Research-related injury care	Sponsor	Stated in the consent form, not only here
Travel and lodging	Sponsor, to a cap	The cap is stated in your consent form
Standard surgical care	Your usual arrangement	Exactly as it would be outside the study
Standard inpatient stay	Your usual arrangement	The 8 to 10 day stay is standard care
Long-term survivorship care	Your usual arrangement	On the routine pathway after handback
Lost income	**Not settled**	This protocol does not compensate it; H. R. 9510 v5 would address it
Continued drug access after closure	**Not settled**	Not guaranteed by this protocol

12.4 After your last visit

[DRAFTING INSTRUCTION] *Roughly 1,000 characters introducing Figure 30. Care is handed back to the participant's own oncology team in writing; a full record copy goes to both the participant and that team; the device support obligation ends at study closure; survivorship follow-up returns to the routine pathway; the data vault remains retrievable for fifteen years; and study results are sent to the participant whether or not they completed. Cite* `natlpaiplatf` *for the post-trial platform proposal.*

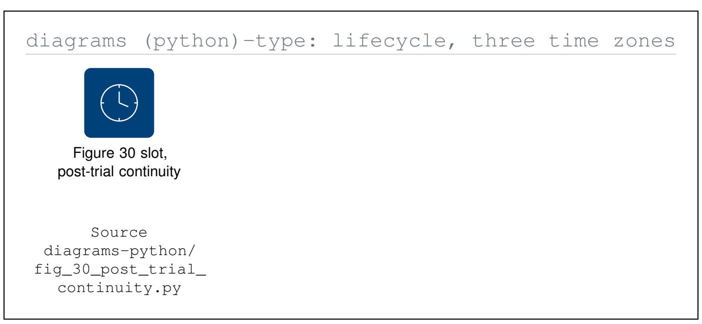

Figure 30. Post-trial continuity across three time zones, during the study, the 12 months after the last visit, and beyond, with the payer attached to every item and the two unsettled items drawn as unsettled rather than quietly left out.

[DRAFTING INSTRUCTION] *Replace the Figure 30 slot with the Diagrams (Python)-type lifecycle from* `diagrams-python/fig_30_post_trial_continuity.py`*. Three* `dgcluster` *time zones plus one* `dgcluster2` *payer container; obligations that end drawn with a 3 mm* `protoblack` *terminator bar at their time-zone boundary; the legislative proposal drawn as a dotted edge, not a solid one, because it is a proposal.*

12.5 Closing argument

[DRAFTING INSTRUCTION] *Roughly 1,400 characters, and write it as an advocate who has just spent twelve sections earning the right to make it. The argument is not that this trial is safe because it is new. It is that a participant who has read this paper knows more about what will happen to them than a participant undergoing a standard Whipple outside a trial: they know the force limits, the stopping budget, the read list on their own data, the name of the person accountable for each failure class, and the two costs the protocol does not cover. Restate the honest gaps rather than closing over them. Cite* `hr9510billv5`, `clintrustpai`, *and* `onpremwhippl`.

13 References and Back Matter

[DRAFTING INSTRUCTION] *Back matter is adapted from `inputs/phase-1-trial-protocol.zip`, `sections/sec-12-references-backmatter.tex`, and re-scoped to an advocacy paper: the abbreviation list keeps only terms this paper actually uses, the amendment history becomes a version history, and a figure inventory is added because thirty figures need an index.*

13.1 Abbreviations

Term	Meaning	Term	Meaning
AE	Adverse event	IRB	Institutional Review Board
CA 19-9	Carbohydrate antigen 19-9	ISGPS	International Study Group on Pancreatic Surgery
CRO	Contract research organisation	LLM	Large language model
DL1 to DL3	Dose levels 1 to 3	OS	Overall survival
DLT	Dose-limiting toxicity	PDAC	Pancreatic ductal adenocarcinoma
DSMB	Data and Safety Monitoring Board	PFS	Progression-free survival
ECOG	Eastern Cooperative Oncology Group	R0	Microscopically clear resection margin
e-stop	Emergency stop	RECIST	Response Evaluation Criteria in Solid Tumors
FDA	Food and Drug Administration	SAE	Serious adverse event
IDE	Investigational Device Exemption	SMV	Superior mesenteric vein
IND	Investigational New Drug	USL	Unified Safety Level

[DRAFTING INSTRUCTION] *Add any abbreviation introduced by `full-patient` that is not in this list, and remove any listed term that the finished paper never uses.*

13.2 Figure inventory

[DRAFTING INSTRUCTION] *Populate all thirty rows from the five stage READMEs. Verify the numbering is continuous from 1 to 30 with no gaps and no duplicates, and that the type counts are Mermaid 9, D2 7, PlantUML 5, Graphviz 5, and Diagrams (Python) 4.*

Fig	Type	§	Subject
1	Mermaid-type	1	The seven commitments made to the participant
2	Graphviz-type	1	Accountability chain rooted at the participant
3	Mermaid-type	2	Journey schema with every decision point marked
4	D2-type	2	Schedule of Activities from the participant's chair
5	D2-type	3	Twenty-one concerns in six disjoint families
6	Graphviz-type	3	Every concern wired to its answering clause
7	Mermaid-type	3	Concern prevalence against answer completeness
8	PlantUML-type	3	Safeguards as capabilities the participant holds
9	Diagrams (Python)	3	Where each concern physically lives
10	Mermaid-type	4	Objective, endpoint, and the sentence it authorises
11	D2-type	4	Endpoint registry with a plain-meaning field
12	PlantUML-type	5	Participant status as a guarded state machine
13	Graphviz-type	5	The 3+3 rule as an auditable decision tree
14	Mermaid-type	6	Eligibility as a two-way gate
15	PlantUML-type	6	Direct booking and sponsor response
16	D2-type	6	The five choices the participant keeps
17	Diagrams (Python)	7	Operating room and on-premises stack
18	PlantUML-type	7	Emergency-stop timing budget
19	Mermaid-type	7	Advise, approve, execute, and log
20	Graphviz-type	7	Hazard fault tree with named barriers
21	D2-type	7	Force caps and vascular no-fly envelope
22	PlantUML-type	8	Four withdrawal routes and their data outcomes
23	Mermaid-type	8	Consent lifecycle with forced re-consent
24	Diagrams (Python)	9	Data pipeline with per-store read lists
25	Mermaid-type	9	Participant visit calendar with totals
26	D2-type	9	Ten robot instruction cards, PDAC-scoped
27	Mermaid-type	10	Five-panel quantitative reassurance dashboard
28	Graphviz-type	10	Evidence provenance for every quoted number
29	D2-type	11	Responsibility matrix, one accountable per row
30	Diagrams (Python)	12	Post-trial continuity and the payer per item

13.3 Version history

Version	Date	Change
Draft 1.0	July 31, 2026	First release. Thirteen sections, thirty figures, twenty-one documented concerns answered, three-stage build from diagram sources through draft and full to final.

Data and code availability

[DRAFTING INSTRUCTION] *Roughly 500 characters. Every figure source, every LaTeX source, and every stage narrative is in the public repository at* `github.com/kevinkawchak/robotic-surgeries/tree/main/patient-robot-advocacy`*. No participant data exists, because no participant exists: this is an independent advocacy paper about a protocol, not a report of a conducted study.*

Acknowledgments

[DRAFTING INSTRUCTION] *Roughly 400 characters. Acknowledge the two dated research passes at* `research/research-a.md` *(Gemini 3.1 Pro) and* `research/research-b.md` *(ChatGPT 5.6 Thinking Extended), and state that the LaTeX build was adapted using Claude Code Opus 5.*

Ethical disclosures

[DRAFTING INSTRUCTION] *Roughly 500 characters. No human participants, no animals, no identifiable data. This is an independent research paper and practical adoption guide, not an active IND or IDE, not an approved protocol, and not medical or regulatory advice. Not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor, CRO, site, regulator, or medical society.*

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Kawchak K. *Patient Robot Advocacy: A Phase 1, First-in-Human, PDAC Clinical Trial Protocol of a LLM-Directed Robotic Whipple with Daraxonrasib (RMC-6236)*. Draft 1.0. Zenodo; 2026. 10.5281/zenodo.xxxxxxxx. Repository v1.0.0: github.com/kevinkawchak/robotic-surgeries/tree/main/patient-robot-advocacy.

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